

Marcel Harrer, M.Sc.

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Professional Summary

AI Engineer with 3+ years of experience building production-grade machine learning systems across computer vision, NLP, time series forecasting, and agent-based architectures. Strong background in applied mathematics (M.Sc.) with hands-on experience in Retrieval-Augmented Generation (RAG), intelligent agents, and scalable data pipelines. Proven ability to translate research into reliable, real-world AI solutions.

Education

University of Vienna

M.Sc. in Applied Mathematics

Oct 2020 – Nov 2023

University of Vienna

B.Sc. in Mathematics

Oct 2016 – Sep 2020

HTL Mödling

Technical High School

Oct 2011 – Jun 2016

Experience

Freelancer — AI Developer

Remote | Mar 2025 – Present

- Designed and deployed intelligent agent systems using LangGraph, including a Text-to-SQL agent enabling natural language access to PostgreSQL databases.
- Built modular tool interfaces supporting structured reasoning, database querying, and automated email workflows.
- Optimized agent orchestration, prompt strategies, and multi-step reasoning pipelines for reliability and accuracy.

ImageTwin — AI Research Engineer

Jan 2024 – Present

- Developed computer vision algorithms to detect image manipulation and integrity violations in scientific figures.
- Implemented keypoint-based image matching using OpenCV (SIFT, SURF) and trained LightGlue models for robust visual correspondence.
- Co-developed a deep learning pipeline for AI-generated image detection, covering data annotation, preprocessing, and automated evaluation.
- Contributed to dataset creation and experimental validation in a research-driven production environment.

Financial Market Authority (FMA) — AI Engineer

Dec 2022 – Dec 2023

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for answering domain-specific regulatory queries using FAISS and transformer-based embeddings.
- Enabled semantic search over large German-language document collections, improving information retrieval accuracy and efficiency.
- Applied SBERT-based similarity models for document clustering and regulatory text analysis.
- Produced analytical dashboards and visualizations in R to support data-driven insights in the Austrian insurance sector.

Projects

Master's Thesis – Accelerating Gradient Descent via L-Smoothness

Developed an adaptive optimization algorithm leveraging L-smoothness to dynamically adjust step sizes in gradient descent. Demonstrated provable convergence guarantees and improved convergence speed compared to fixed-step methods.

Agent Lawyer

Built a legal assistant system using Retrieval-Augmented Generation (RAG) to provide grounded, document-based answers to legal queries. Implemented semantic search with FAISS, transformer embeddings, and a reranking model to improve answer relevance. Runs locally on a compact LLaMA 3.2 model.

GitHub: github.com/MarcelHarrer/agent-lawyer-public

Skin Cancer Detection with CNNs

Trained convolutional neural networks on the HAM10000 dataset for skin lesion classification. Implemented preprocessing, data augmentation, and model evaluation pipelines, achieving performance comparable to expert-level benchmarks.

Technical Skills

Languages: Python, SQL, R, MATLAB

ML / AI: PyTorch, TensorFlow, Hugging Face Transformers, Computer Vision, Deep Learning, Time Series Forecasting, Retrieval-Augmented Generation (RAG), Intelligent Agents

Backend & MLOps: FastAPI, Docker, CI/CD, FAISS

Cloud & AWS: AWS SageMaker, AWS S3, AWS Lambda, AWS EC2, AWS Bedrock, AWS CloudWatch, AWS IAM

Data & Analytics: Pandas, Power BI, SQL Databases

Tools: Git, LangChain, LangGraph, OpenCV

Mathematics: Optimization, Applied Mathematics, Data Science

Languages (Spoken): German (Native), English (Bilingual), Spanish (Advanced)